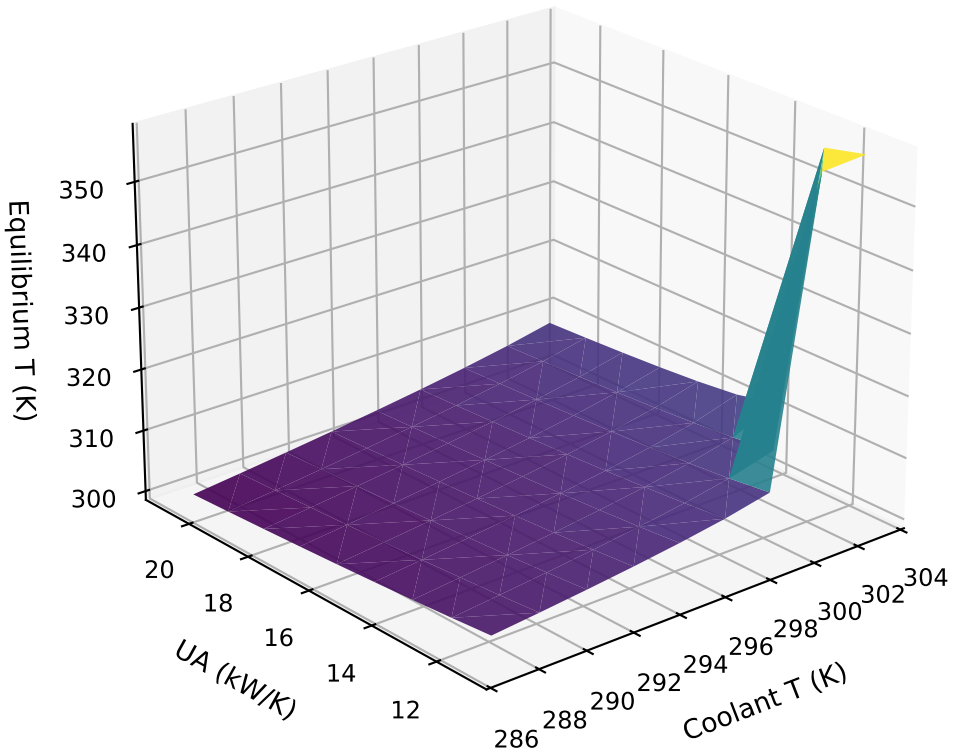Lowest stable equilibrium in each scenario: branch transitions